

Table 6.1 Returns of Asset Classes in 2008

Cash	Three-month T-bill	1.3%
Core Bonds	Barcap Aggregate Index	5.2%
Global Bonds	Citigroup World Government	10.9%
TIPS	Citigroup US Inflation Linked	−1.2%
Emerging Market Bonds	JPM Emerging Markets Bond Index	−9.7%
US High Yield	Merrill Lynch High Yield Master	−26.3%
Large Cap Equity	S&P 500	−37.0%
Small Cap Equity	Russell 2000	−33.8%
International Equity	MSCI World ex US	−43.2%
Emerging Markets Equity	IFC Emerging Markets	−53.2%
Public Real Estate	NAREIT Equity REITS	−37.7%
Private Real Estate	NCREIF Property Index	−16.9%
Private Capital	Venture Economics (Venture and Buyouts)	−20.0%
Equity Hedge Funds	HFRI Equity Hedge Index	−20.6%
Fixed Income Hedge Funds	HFRI Fixed Income Index	−17.8%
Commodities	Dow Jones AIG Commodity Index	−35.7%

2. Factor Theory

Factors are to assets what nutrients are to food. Table 6.2 is from the Food and Nutrition Board, which is part of the Institute of Medicine of the National Academies, and lists recommended intakes of the five macronutrients—water, carbohydrates, protein, fiber, and fat—for an “average” male, female, and child. Carbohydrates can be obtained from food made from cereals and grains. Protein is obtained from meat and dairy products. Fiber is available from wheat and rice. Fat we can consume from animals but also certain plant foods such as peanuts. Each type of food is a bundle of nutrients. Many foods contain more than just one macronutrient: for example, rice contains both carbohydrates and fiber. Different individuals, whether sick or healthy, male or female, or young or old, have different macronutrient requirements. We eat food for the underlying nutrients; it is the nutrients that give sustenance.

Factor risks are the driving force behind assets’ risk premiums. An important theory of factor risk is the CAPM, which we explore in the next section. The CAPM